Experiment 2
Perform Operations on Vectors, Matrices and Arrays in R

PART B

PART B: TO BE COMPLETED BY STUDENTS)

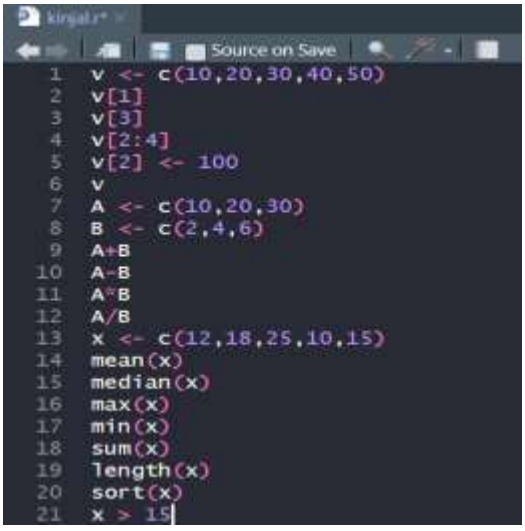
(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no ERP access available)

Roll No. 60	Name: Avdhut Parish Madhavi
Class : S.E. - C	Batch : C-2
Date of Experiment:	Date of Submission:-
Grade :	

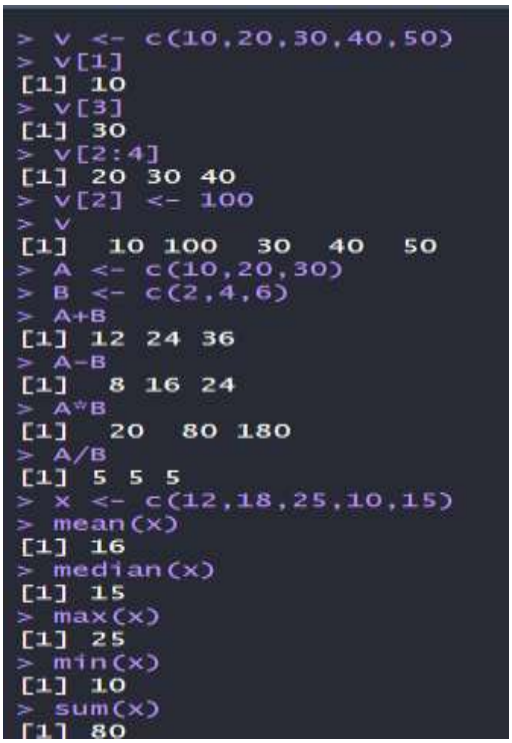
Observation: (Run the programs given above in R studio and paste screenshots of along with the output)

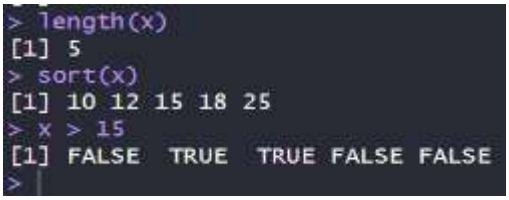
1. **Vector Operations**

   Input :-

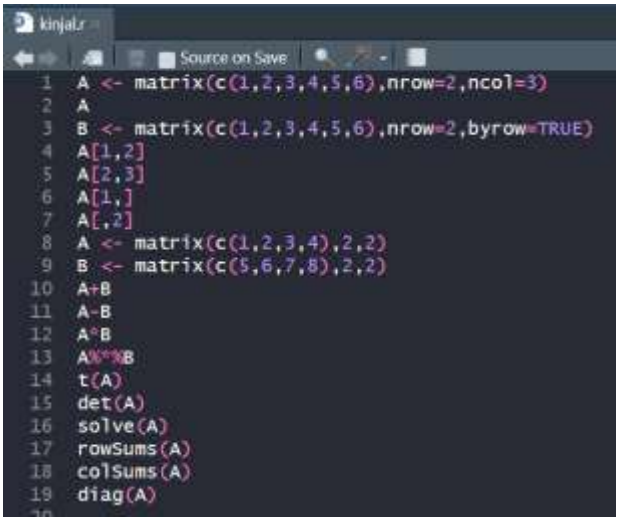


Output:-





2. **Matrix Operations**
   Input:-



Output:-

```
> A <- matrix(c(1,2,3,4,5,6),nrow=2,ncol=3)
> A
     [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6
> B <- matrix(c(1,2,3,4,5,6),nrow=2,byrow=TRUE)
> A[1,2]
[1] 3
> A[2,3]
[1] 6
> A[1,]
[1] 1 3 5
> A[,2]
[1] 3 4
> A <- matrix(c(1,2,3,4),2,2)
> B <- matrix(c(5,6,7,8),2,2)
> A+B
     [,1] [,2]
[1,]    6   10
[2,]    8   12
> A-B
     [,1] [,2]
[1,]   -4   -4
[2,]   -4   -4
> A*B
     [,1] [,2]
[1,]    5   21
[2,]   12   32
```

```
> A%*%B
     [,1] [,2]
[1,]   23   31
[2,]   34   46
> t(A)
     [,1] [,2]
[1,]    1    2
[2,]    3    4
> det(A)
[1] -2
> solve(A)
     [,1] [,2]
[1,]   -2  1.5
[2,]    1 -0.5
> rowSums(A)
[1] 4 6
> colSums(A)
[1] 3 7
> diag(A)
[1] 1 4
>
```

3. **Array Operations**
   Input:-

```
A <- array(1:24,dim=c(3,4,2))
A[2,3,1]

A[1,,2]

A[,,1]
A1 <- array(1:12,dim=c(2,3,2))

A2 <- array(13:24,dim=c(2,3,2))

A1+A2

A1*A2
dim(A)
length(A)
apply(A,c(1),sum)
apply(A,c(2),mean)
```

Output:-

```
> A <- array(1:24,dim=c(3,4,2))
> A[2,3,1]
[1] 8
> A[1,,2]
[1] 13 16 19 22
> A[,,1]
     [,1] [,2] [,3] [,4]
[1,]    1    4    7   10
[2,]    2    5    8   11
[3,]    3    6    9   12
> A1 <- array(1:12,dim=c(2,3,2))
> A2 <- array(13:24,dim=c(2,3,2))
> A1+A2
, , 1

     [,1] [,2] [,3]
[1,]   14   18   22
[2,]   16   20   24

, , 2

     [,1] [,2] [,3]
[1,]   26   30   34
[2,]   28   32   36
```

```
> A1*A2
, , 1

     [,1] [,2] [,3]
[1,]   13   45   85
[2,]   28   64  108

, , 2

     [,1] [,2] [,3]
[1,]  133  189  253
[2,]  160  220  288

> dim(A)
[1] 3 4 2
> length(A)
[1] 24
> apply(A,c(1),sum)
[1]  92 100 108
> apply(A,c(2),mean)
[1]  8 11 14 17
>
```

Observation Table:

Operation	Result
Vector Creation	Successfully created one-dimensional vectors
Vector Arithmetic	Addition, subtraction, multiplication, and division performed correctly

Operation	Result
Statistical Functions	Mean, median, sum, maximum, and minimum computed successfully
Matrix Operations	Matrix addition, multiplication, transpose, determinant, and inverse executed correctly
Array Operations	Three-dimensional arrays created and manipulated successfully
Apply Function	Row-wise and column-wise computations performed successfully

Result

The experiment was successfully performed. Vectors, matrices, and arrays were created using R, and various arithmetic, statistical, and matrix operations were executed correctly. The experiment demonstrated efficient manipulation of one-dimensional, two-dimensional, and multidimensional data structures in R.

Applications

- Statistical data analysis
- Scientific computing
- Machine Learning preprocessing
- Image processing (multidimensional arrays)
- Financial data analysis
- Engineering computations
- Healthcare analytics
- Artificial Intelligence applications

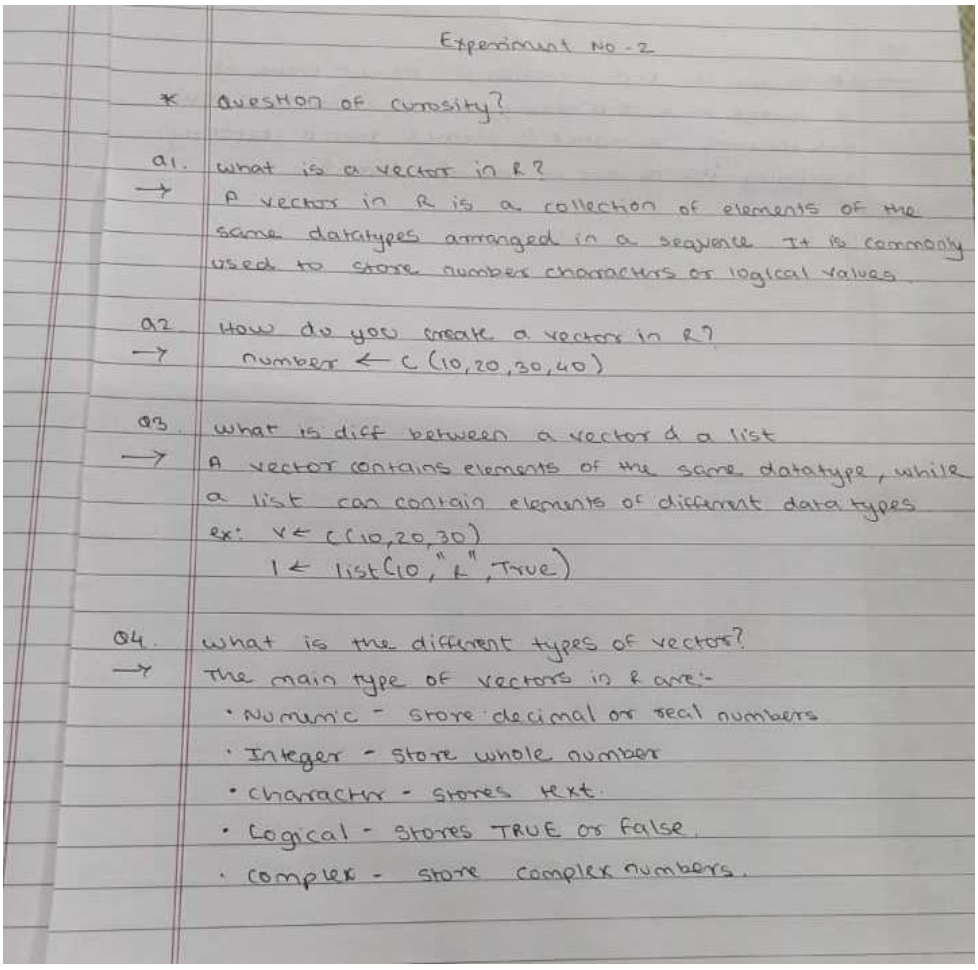
Precautions

1. All elements in a vector or matrix must be of the same data type.
2. Ensure matrices have compatible dimensions before performing matrix multiplication.
3. The inverse of a matrix exists only if its determinant is non-zero.
4. Use proper indexing while accessing vector, matrix, or array elements.
5. Verify dimensions before performing arithmetic operations on arrays.
6. Save the R script before execution to avoid loss of work.

Questions of Curiosity

1. What is a vector in R?
2. How do you create a vector in R?
3. What is the difference between a vector and a list?
4. What are the different types of vectors?
5. How is a matrix different from a vector?

Answers:-



Experiment No - 2

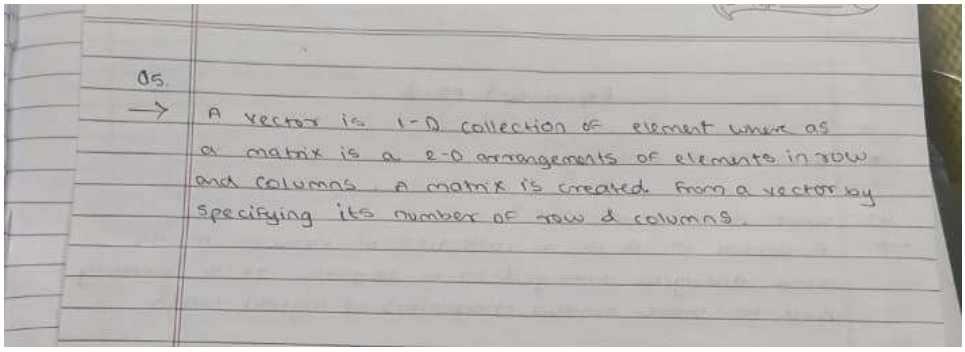
* Question of curosity?

Q1. What is a vector in R?
→ A vector in R is a collection of elements of the same datatypes arranged in a seqvence. It is commonly used to store numbers characters or logical values.

Q2 How do you create a vector in R?
→ number ← c (10,20,30,40)

Q3. What is diff between a vector & a list
→ A vector contains elements of the same datatype, while a list can contain elements of different data types
ex: v ← c(10,20,30)
l ← list(10, "k", True)

Q4. What is the different types of vectors?
→ The main type of vectors in R are:-
- Numeric - store decimal or real numbers
- Integer - store whole number
- character - stores text.
- Logical - stores TRUE or false.
- complex - store complex numbers.



Viva Voce Questions

6. What is a vector in R?
7. How do you create a vector in R?
8. What is the difference between a vector and a list?
9. What are the different types of vectors?
10. How is a matrix different from a vector?
11. Explain the `matrix()` function.
12. What is the purpose of `%*%` in R?
13. Differentiate between `*` and `%*%`.
14. What is the transpose of a matrix?
15. What is the determinant of a matrix?
16. When can the inverse of a matrix be calculated?
17. What is an array in R?
18. How is an array different from a matrix?
19. Explain the `apply()` function.
20. What is the purpose of `dim()` and `length()` functions?
21. How do you access elements in a three-dimensional array?
22. What statistical functions are commonly used with vectors?
23. What happens if vectors of unequal lengths are added in R?
24. What are the applications of matrices in data science?
25. Why are vectors, matrices, and arrays important in R programming?